INCH-POUND

MIL-G-44439
30 September 1992

MILITARY SPECIFICATION

GINGERBREAD, POWDER

(FOR ORAL LIQUID FEEDING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gingerbread powder, unit packed in pouches or cans, for oral liquid feeding of hospitalized patients. This product is one component of a meal that, when consumed with other components in three meals or supplementary nourishments, provides a nutritionally adequate, orally consumed diet (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

MILITARY

MIL-P-44151 - Packaging into Pouches and Meal Assembly of Oral Liquid Feeding Powders

MIL-M-44458 - Meal Module, 10-Patient, Packaging of Oral Liquid Feeding Powders into Cans and Assembly of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Dry Whey (7 CFR Part 58)
U.S. Standards for Instant Nonfat Dry Milk (7 CFR Part 58)
General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58)

(Copies are available from the Dairy Standardization Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Room 2750, South Building, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the

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issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copie should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Sugar, brown, granulated. The granulated brown sugar shall be a granular, free-flowing sugar, whose granules break apart in solution. It shall have a medium golden brown color, and a maximum moisture content of 2.3 percent. It shall have a pH of 6.0.

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3.2.2 Milk, nonfat, dry, instant. The instant, nonfat, dry milk shall be U.S. Extra Grade as defined in the United States Standards for Instant Nonfat Dry Milk (7 CFR Part 58).

3.2.3 Creaming agent, non-dairy. The non-dairy creaming agent shall be a cold water dispersible creaming agent. It shall possess a white to cream color, shall be free of hard lumps, and 100 percent shall pass through a U.S. Standard No. 20 sieve. The non-dairy creaming agent shall contain 65 to 75 percent fat and a moisture content that does not exceed 4.0 percent (see 6.4.1). The non-dairy creaming agent shall be manufactured in a USDA approved plant that meets the requirements specified in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58).

3.2.4 Molasses powder. The molasses powder shall be spray-dried factory molasses containing 60 percent molasses solids. It shall be dark brown in color, and have the odor and flavor typical of molasses. It shall have a 4.0 percent maximum moisture content, and 100 percent shall pass through a U.S. Standard No. 20 sieve (see 6.4.2).

3.2.5 Whey, dry, sweet-type. The dry, sweet-type whey shall be U.S. Extra Grade as defined in the United States Standards for Dry Whey (7 CFR Part 58).

3.2.6 Starch, high viscosity, precooked, modified, tapioca. The starch shall be a pure, off-white, and odorless, slightly coarse-ground powder. The starch shall have a maximum moisture of 9.0 percent and a pH of 5.0 to 7.0. The starch shall be bland in taste, have excellent low temperature stability, and shall be rapidly dispersed with moderate agitation.

3.2.7 Cinnamon, ground. Ground cinnamon shall be the dried bark of Cinnamomum burannii Blume commonly known as "Korintji Cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.8 Flour, wheat. Wheat flour shall be free-flowing, enriched, general purpose white (bleached).

3.2.9 Ginger, ground. Ground ginger shall be derived from the root or rhizome of Zingiber officinale Roscoe and shall possess an aromatic odor and flavor with a tan to pale brown color. The ground ginger shall contain not less than 1.5 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 35 sieve.

3.2.10 Nutmeg, ground. Ground nutmeg shall be derived from the dried kernel of Myristica fragrans Houtt. The nutmeg shall have a dull brown color. The ground nutmeg shall contain not less than 6.5 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

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3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of the gingerbread powder. The gingerbread powder shall be formulated and prepared as follows:

Ingredients	Percent by weight
Sugar, brown, granulated	37.93
Milk, nonfat, dry, instant	21.50
Creaming agent, non-dairy	13.00
Molasses powder	11.28
Whey, dry, sweet-type	9.43
Starch, high viscosity, precooked, modified tapioca	3.41
Cinnamon, ground	1.88
Flour, wheat	1.00
Ginger, ground	0.38
Nutmeg, ground	0.19

a. All ingredients shall be combined and uniformly blended in a suitable mixer until no lumps exist, and the product is homogeneous.

b. The product shall be handled in such a manner so as to prevent an increase in moisture content (see 3.4.1) and completely packaged and sealed as to meet finished product requirements.

3.3.2 Pouch filling and sealing. Each pouch shall be filled and sealed in accordance with the filling and sealing requirements specified in MIL-P-44151. Each pouch shall be filled to comply with the finished product requirements.

3.3.3 Can filling and sealing. Each can shall be filled and sealed in accordance with the filling and sealing requirements specified in MIL-M-44458. Each can shall be filled to comply with the finished product requirements.

3.4 Finished product requirements.

3.4.1 Dry product. The dry product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

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d. The product shall be free from caking and lumps which do not fall apart under light pressure.

e. One hundred percent of the product shall pass through a U.S. Standard No. 7 sieve.

f. The moisture content shall not exceed 5.0 percent.

g. The aerobic plate count shall not be greater than 25,000 per gram in four of five samples and not greater than 50,000 per gram in any individual sample.

h. The Escherichia coli count shall be less than 1 per gram (no positives in the standard 3 tube MPN technique).

i. The Salmonella test shall be negative per 25 grams of product.

j. The net weight of each individual pouch shall be not less than 95 grams and not greater than 105 grams.

k. The lot average net weight of the pouch shall be not less than 100 grams.

l. The net weight of each 401 x 602 can shall be not less than 475 grams.

m. The lot average net weight of the 401 x 602 can shall be not less than 500 grams.

3.4.2 Hydrated product. Hydration of the dry powder shall be in accordance with the specified instructions on the pouch or can label, as applicable. The hydrated product shall comply with the following requirements:

a. After hydration for oral liquid feeding, no more than 0.5 percent of the product shall remain when the product has passed through a U.S. Standard No. 20 sieve.

b. The hydrated product shall have an odor, color and flavor that is characteristic of gingerbread.

c. For oral liquid feeding, the hydrated product shall have a consistency that allows it to be easily sipped through a drinking tube.

3.4.3 Palatability. The hydrated product shall be equal to or better than the approved preproduction sample (see 6.2) in palatability and overall appearance.

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3.5 Plant qualifications. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each

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ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44151.

4.5.4 Filled and sealed can examination. Examination of the filled and sealed cans shall be in accordance with the quality assurance provisions of MIL-M-44458.

4.5.5 Net weight examination. The filled and sealed pouches or cans shall be examined for net weight in accordance with 4.5.5.1 or 4.5.5.2.

4.5.5.1 Net weight examination of pouches. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 95 grams or greater than 105 grams shall be classified as a minor defect. Results shall be reported to the nearest gram. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 100 grams.

4.5.5.2 Net weight examination of cans. The net weight of the filled and sealed cans shall be determined by weighing each sample unit on a suitable scale tared with a representative empty can and lid. Any individual net weight less than 475 grams shall be classified as a minor defect. The results shall be reported to the nearest gram. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 500 grams.

4.5.6 Product examination. Eight filled and sealed pouches or cans shall be selected at random from the lot and examined for the defects listed in table I. The finding of any major defect shall be cause for rejection of the lot.

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TABLE I. Product defects 1/ 2/

Category	Defect
Major	Dry product
101	Product is not free from caking and lumps which do not fall apart under light pressure
102	Any product remaining on a U.S. Standard No. 7 sieve 3/
	Hydrated product 4/ 5/
103	More than 0.5 percent of product does not pass through a U.S. Standard No. 20 sieve 6/
104	The hydrated product does not have the odor, color, or flavor of gingerbread
105	The product does not have a consistency which allows it to be easily sipped through a drinking tube 7/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.4.3).

3/ The powder shall be poured onto a U.S. Standard No. 7 sieve and vigorously shaken.

4/ Hydration of the dry powder from the pouch or can shall be accomplished in accordance with the pouch or can label instructions in MIL-P-44151 or MIL-M-44458, as applicable.

5/ After hydration of the required quantity from each sample unit, divide the quantity into two equal parts. Use one part for examination of defect 103 and one part for examination of defects 104 and 105.

6/ The hydrated product shall be poured onto a tared U.S. Standard No. 20 sieve. The product shall be washed for 1-1/2 minutes with slowly running, lukewarm water. The sieve shall be allowed to drain for two minutes, then excess water shall be wiped from the bottom and sides of the sieve. The remaining solids shall be weighed on the tared sieve.

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7/ The product shall be easily sipped through a white or clear plastic drinking tube, 7-3/4 inches long and a 1/4 inch diameter (inside).

4.5.7 Moisture content testing. Eight filled and sealed pouches or cans shall be selected at random from the lot regardless of lot size. The pouched or canned product shall be individually tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 945.43, with the exception that the product shall be dried for 16 hours and no sand or water added. The test results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Microbiological testing. Five filled and sealed pouches or cans shall be selected at random from the lot regardless of lot size. The pouched or canned product shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the AOAC, method 966.23. The diluent shall be added to each sample and allowed to stand for 15 minutes before blending the sample. Salmonella testing shall be in accordance with the Official Methods of Analysis of the AOAC, method 967.26. Any result not conforming to the microbiological requirements in 3.4.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-P-44151 or MIL-M-44458, as applicable.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44151 or MIL-M-44458, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This product is intended to be used by patients requiring an oral liquid or pureed diet (for example, those with maxillofacial injuries, wired jaws, oral surgery, dental disease, or swallowing difficulties).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.3).

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d. Provisions for approved preproduction samples (see 3.4.3 and 6.3).
e. Type of unit pack required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Ingredient information.

6.4.1 Creaming agent, non-dairy. It has been found that San-A-Creme CWS 881 manufactured by Beatreme Foods, Inc., meets the requirements of 3.2.3 and performs satisfactorily in this product.

6.4.2 Molasses powder. It has been found that No. 2399 Molasses Powder manufactured by Borden, Inc., meets the requirements of 3.2.4 and performs satisfactorily in this product.

6.5 Subject term (key word) listing.

Canned food
Combat field feeding
Hospital
Jaw injury
Liquid diet
Operational ration
Oral liquid feeding
Pouch food
Shelf stable

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-A730)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-44439	1992 September 30

3. DOCUMENT TITLE
GINGERBREAD, POWDER (FOR ORAL LIQUID FEEDING)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP4, JUL 91 *Previous editions are obsolete.*

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